ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I B.V. Tharun [192525155] (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tharun B.V

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

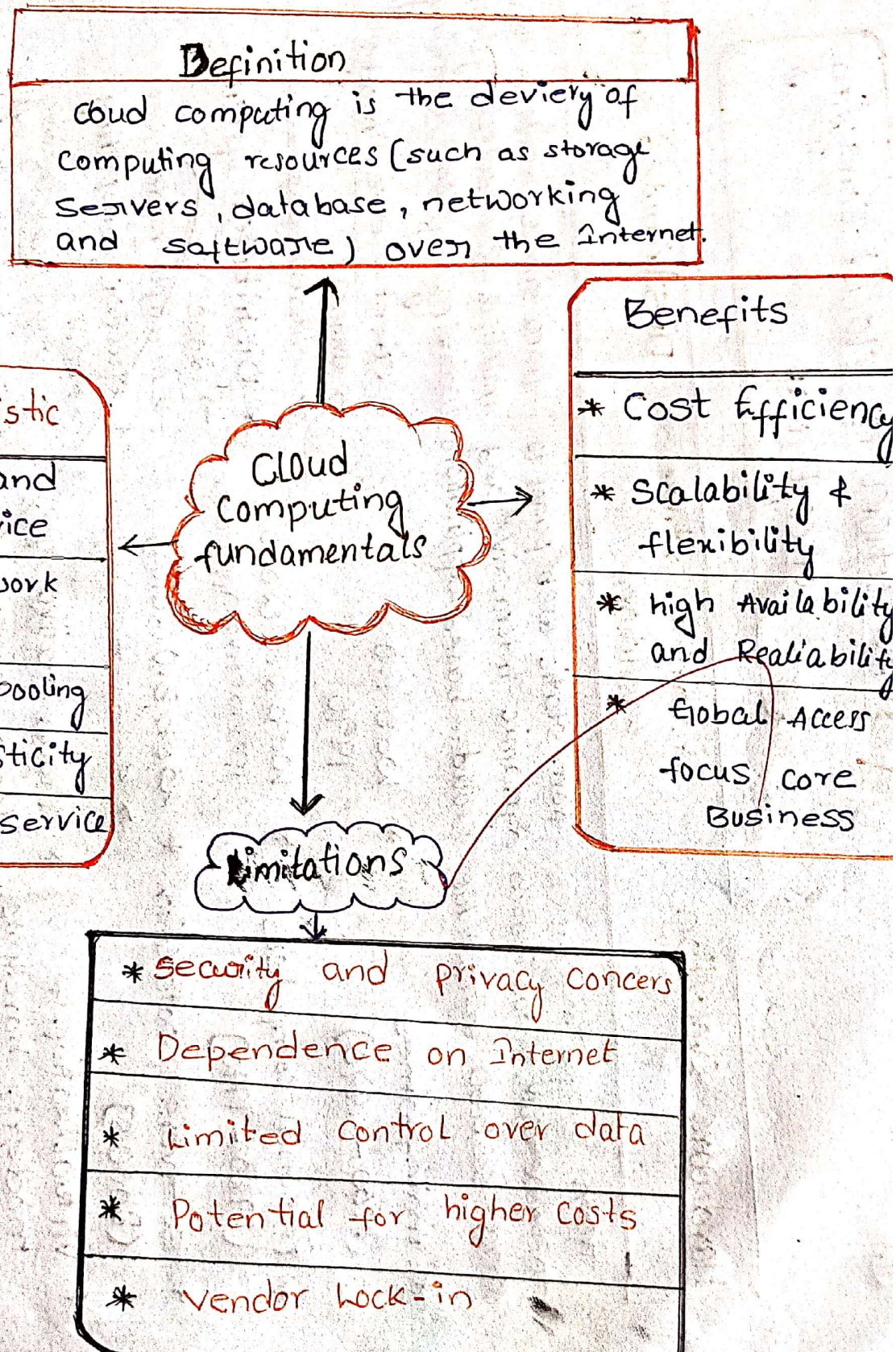
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

1)

Cloud computing -fundamentals

concept map



(2)

Evolution of cloud computing Concept map

Hardware Evolution	Internet software Evolution	Distributed system	service deli
from single machines to powerful, virtualization and scalable infrastruc * mainframes (1960 – 1970) * Pesonal computers (1980 – 1990) * Servers & Data Center (2000s) * virtualization & modern hardware (2010 – Present)	from static web pages to dynamic interactive and Cloud – based applications * Web 1.0 → 1990 * Web 2.0 (2000s) → interactive application * Web 3.0 (2010s) → semantic web * Cloud-Native APPS (2010 – present)	from single servers to multiple connected system working systems working * Client server model → (1990s) → centralized servers * Distributed system → (2000s) → multiple nodes * cluster competing → 2010 → parallel processing * Internet scale system 2010 → Prasent	from infrastructure ownership to on demand, pay-as-you go services. * IaaS → virtual machines → AWS * PaaS → Developoment platf → Example :- google Ap * SaaS → Ready-to-uses appli → gmail, salesforce * Xaas → Extended servics

Enabled by
* Advanced hardware
* internet software
* Distributed system

→ Cloud Computing
The Result
On-demand . Scalable

Lead to
* Resource optimization
* Cost Efficiency

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(3)

Service Models

Iaas, Paas, saas xaas

Iaas

Provides virtualized computing resources over the internet

Ex:- * virtual machines
* storage
* networks

Provides Responsibilities

* Physical data centeres and hardware
* virtualization layer
* networking, storage and computing Infrastructure
* security, availability

User Responsibilities

* operating systems
* Application and middleware
* Data management

Paas (platform as service)

Provides a computete development and runtime environment for building, testing and deploying applications

Provide & Responsibilities

* Infrastructure hardware, networking
* Operating system
* Runtime, middleware development tool
* security, availability

user Responsibilities

* Application code
* Data and database
* Business Logic

Saas

Provides a computet software application over the internet

Ex:- G-mail, salesfoa

Provide & Responsibilities

* Complete application
* Infrastructure and platform
* Data storage and backup
* security, updates
* Availibility and performing

User Responsibilities

* use the applitions
* manage use accounts
* Enscore data security

xaas

A broader n
that includes
service del
over the cl
Ex:- Daas, f
Aiaas, s

Provider

* Deliver the
Coud serv
* infrastru
Security a
Scalability
* Service
* SLA and s

User Respons

* Consume of s
* manage data
* Ensure prope

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(4)

Virtualization and web services as enablers of clod

virtualization

* Creates multiplies virtual machines(VMS] on a singles physical server
* Improves resource utilization and isolation
* Enables flexible allocation of computing resources

Provides isolated and flexible infrastructure

Web Services

* Uses standard protocols (HTTP, ReST, SOAP]
* Allows different applications and system to communicate
* Enable easy Integration and Interoperability

Enable Commonication and integration

Cloud Deployment

* virtualization provides the flexible on-demand infrastructure to run cloud services
* web services enable apps and servies to work togetor

13/10/26

Scalability

* Easy to add or remove resource
* supports rapid scaling and elastic growth

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